

SpeakATX: Technical Report

Overview

[SpeakATX](#) is a website made using React, Vite, Bootstrap, and Node.js. Its goal is to provide resources to minimal English and non-English speakers in Austin, TX. It currently has 15 unique pages.

- The splash page contains a brief overview of SpeakATX's purpose, and some basic navigation.
 - This page is linked in our navigation bar at the top of the site.
- The about page reiterates SpeakATX's purpose and provides information about the developers.
 - This page is linked in our navigation bar at the top of the site.
- There is one page each for translation services, language communities, and job postings.
 - Each of these pages are linked in our navigation bar at the top of the site.
- There are additional pages for each individual instance of services, communities, and job postings.
 - These are linked as cards in their respective parent pages.

About Page

The about page reiterates SpeakATX's purpose and provides information about the developers. For each developer this includes their name, a picture, and a short bio.

The about page also tracks git commits and issues for each developer. This is updated dynamically by fetching data from GitLab's API.

- For both commits and issues, GET requests are made per page of 100 items, looping through pages until there is no data left.
- The resulting commits/issues are then mapped to their authors/assignees and counted.

Models, Instances, and Attributes

SpeakATX has 3 core models: services, communities, and job postings. Each model contains some number of instances. Each instance has several attributes.

Each model page displays the number of instances and a card for each instance. Each card contains an image and lists the instance attributes. These cards serve as clickable links which lead to their individual instance pages.

Instance pages for each model contain further information and a button that links to the external website for that instance. Varying by model, some instances may additionally feature images and maps. We have added a back button to each instance page for convenience, as we expect users to browse and click through multiple pages.

Instances also connect to related instances of other models. Currently, SpeakATX will display cards at the bottom of the page for related instances that match the languages of the current instance.

- Language is an important attribute in SpeakATX. All instances have at least one language.
- Instances can have multiple languages.

Related instances are found by iterating through the lists of instances that belong to the other models and selecting those that have at least one matching language.

We store attribute information for all instances of all models in the instances.jsx file as static arrays belonging to the Instances class. Currently, they have been manually inputted. For the next phase, they will be dynamically generated based on data fetched from our data sources. The attributes are as follows:

- Translation service attributes:
 - Name (string)
 - Rating (int 1-5)
 - Area of Austin (string)
 - Price (int)
 - Languages (string)
- Job attributes:
 - Name (string)
 - Title (string)
 - Hourly Pay (int)
 - Area of Austin (string)
 - Languages (string)
- Community attributes:
 - Name (string)
 - Area of Austin (string)
 - Community type (string)
 - Number of members (int)
 - Languages (string)

User Stories

We implemented four of our customer's user stories as requested, which were:

- Adding a translation feature on the splash page
- Adding a pie chart that displays the breakdown of foreign languages spoken in Austin
- Displaying the price of translation services on the cards instead of the budget range
- Fixing the layout on the About page so that the bios do not overlap with the team members' photos

We negotiated a user story that suggested we put services directly on the splash page because we felt that it wasn't fitting to put instances of a model on it. However, we did change the splash page's layout slightly since our customer felt that it looked empty.

Hosting

SpeakATX is hosted on AWS. The domain name is registered on Namecheap.

API

SpeakATX will use a custom RESTful API.

- Documentation: <https://documenter.getpostman.com/view/42364456/2sAYXFiHS8#intro>

Challenges

In developing this portion of SpeakATX, we ran into some minor challenges. Most were simply due to the learning curve, as several of us were new to web development. Some examples include properly centering UI elements and getting images to load correctly. These challenges were overcome through reading documentation, browsing internet forums, and consulting LLMs such as ChatGPT.